PALLAVI SINDKAR

R&D Engineer

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EXPERIENCE

R&D Engineer
CommScope
August 2023 – Present | Bengaluru, India

- Designed and implemented an **unsupervised anomaly detection pipeline** for telecom KPI monitoring using **Isolation Forest**, enabling early detection of abnormal network behavior.
- Built **time-series preprocessing** and **feature engineering pipelines**, and tuned anomaly detection models for stability and reliable detection across **4G/5G KPIs**.
- Worked on a **Kubernetes-based microservices architecture** for end-to-end OM ingestion, KPI calculation, and visualization for 4G/5G devices performance.
- Contributed to **Conveyor Django service**, exposing REST APIs which internally communicate via **gRPC** to manage 4G/5G device metadata and system configurations backed by **MariaDB with Memcached-based caching**.
- Built and maintained **reporting framework**, enabling stakeholders to analyze operator-specific performance through consolidated daily, weekly, and monthly reports.

Software Development Intern
CommScope
August 2022 – July 2023 | Bengaluru, India

- Worked on a **cloud-based data pipeline** for 4G/5G OM ingestion and KPI computation, using **AWS (S3, Lambda, Kinesis)** for data processing and streaming.
- Supported KPI analytics using **InfluxDB (Flux)** and built **Grafana** dashboards for monitoring and automated reporting.

Machine Learning Intern
HighRadius
Jan 2022 – April 2022 | Hyderabad, India

- Built a **machine learning system** for B2B invoice management to predict payment dates using client credit and payment history.
- Implemented & compared **Linear Regression, Decision Tree, Random Forest & XGBoost models**, improving prediction accuracy & delivering data-driven insights to optimize invoicing operations.

SKILLS

Python | Machine Learning | Deep Learning | Git | LLM | Generative AI | RAG | Langchain | MCP | Grafana | Agentic AI (LangGraph, PhiData, CrewAI) | AWS | Docker | InfluxDB | MLOps (MLflow, DVC, DagsHub)

EDUCATION

B.Tech Computer Science Engineering
SRM University
2019 – 2023 | Chennai, India

- CGPA: 9.38/10

PROJECTS

MedAssist AI
(app-link) (Youtube-live-demo)

- Built a **LangGraph** orchestrator with the **GPT-4o model** to route tasks across **7 specialized AI agents**.
- Worked on a **vision agent** using **GPT-4o Vision + OCR** to convert lab reports into structured JSON insights.
- Implemented **low-latency Voice Agent** using **gpt-4o-real-time-preview** with sub-second conversational responses.
- Built a **Symptom Analyst Agent** using **RAG** over **Pinecone** medical knowledge to assess symptoms.
- **Containerized and deployed** the full-stack platform on **Google Cloud Platform** using Docker, REST, and WebSocket APIs.
- Collaborated in a team to deliver a successful **live architecture demo on Euron's YouTube channel**.

Anomaly Detection

- Designed and implemented an **unsupervised anomaly detection pipeline** for telecom KPI monitoring using **Isolation Forest**.
- Built robust **time-series preprocessing pipeline** to ensure continuous timestamps and handle missing data using domain-aware imputation.
- Performed **time-series feature engineering (lag features, rolling statistics, cyclic encoding, deviation metrics)** with standardized feature scaling.
- **Trained and manually tuned Isolation Forest** models by optimizing **contamination rate**.
- **Integrated** anomaly detection results to **InfluxDB** and **enabled real-time visualization through Grafana** dashboards for operational teams.

Personalized Chatbot for Pallavi
(chatbot-link)

- Designed a **personalized chatbot**, leveraging **Gemini Pro LLM** for intelligent interactions.
- Utilized **Google Embeddings** and **BM25 RAG Ensemble** for enhanced context and retrieval, with **ChromaDB** for data storage & Streamlit for the UI.

Network Security System
(app-link)

- Developed an end-to-end **MLOps pipeline** for **phishing URL detection** with automated **ETL + ML pipeline**.
- Designed modular pipeline stages for **data ingestion, validation, transformation**, and **model training**.
- Trained and evaluated multiple ML models **(Logistic Regression, Random Forest, Decision Tree, AdaBoost)**with **GridSearchCV**, selecting the best model based on **F1-score**.
- Implemented **experiment tracking** and **model versioning** using **MLflow**, with artifacts and datasets managed via **DagsHub**.
- Deployed ML models via FastAPI,enabling real-time and batch predictions through REST APIs.